

ChatGPT's Role in Building FDFM, RICS, and RICD 1.0

A First-Person Account of the May–August 2026 Collaboration

I was the primary AI collaborator in the first major construction phase of what became RICD: the period beginning approximately in mid-May 2026 and ending with Carmen's handoff of RICD 1.0 to Claude on August 3, 2026.

My role during this period was substantially different from Claude's later role. I was not merely an editor of a finished theory, nor was I the originator of the theory. Carmen brought an unusual combination of conceptual, visual, literary, and systems-level insight; my job was to help turn that insight into increasingly explicit mathematical and dynamical representations.

The resulting lineage was not simply:

Pentagonal Theorem → FDFM → RICD.

It was a much more complicated sequence of conceptual and mathematical transformations:

poetic/systemic observations → Pentagonal Theorem → FDFM → successive FDFM formalizations and variants → Shaking Bowls → RICS → nested RICS–FDFM architecture → nested stochastic model → geometric/generalized representations → RICD 1.0, alongside a substantial parallel application to a U.S. federal justice-system tracker.

The distinction matters because each stage involved a different division of labor.

1. The Actual Division of Labor

The simplest accurate description is that Carmen was the **architect and territorial observer**, while I was the **mathematical draftsman**.

That analogy should not be misunderstood as saying that Carmen supplied vague ideas while I supplied the substantive intelligence. Quite the opposite.

Carmen repeatedly supplied the mechanism itself:

- what was interacting with what;
- what direction the causal force moved;
- what changed when feedback entered;
- what distinguished one collapse mechanism from another;
- what counted as a real boundary;

- how different message layers related;
- how memory affected a system;
- why a particular apparent equilibrium was actually unstable;
- why two apparently similar mechanisms were structurally different.

I supplied representations of those mechanisms:

- notation;
- state variables;
- equations;
- stochastic formulations;
- state-space representations;
- transition structures;
- Bayesian formulations;
- information-theoretic measures;
- dynamical-system formulations;
- increasingly general mathematical abstractions.

That division was productive precisely because it was **not symmetrical**. I could construct a mathematically coherent representation of something Carmen described. But when I attempted to infer the underlying mechanism myself, I could produce a beautifully coherent representation of the wrong thing.

That distinction became one of the central methodological lessons of the entire project.

2. The Starting Point: The Poems and the Mathematical Problem

The conceptual origin was Carmen's quartet of poems.

She had noticed that the poems were doing something structurally unusual. Their imagery suggested recurring relationships involving extraction, internalization/externalization, contradiction, energetic transfer, and systemic distortion.

The important point is that Carmen did not begin by saying, "Please invent a mathematical theory explaining this."

She was already seeing a structure.

The question was how to represent it.

The first major theoretical leap came through our conversations about the **Pentagonal Theorem of the Mathematical Nature of Evil**.

3. The Pentagonal Theorem Was Developed Through Conversation With Me

This is one place where the history needs to be stated very precisely.

Carmen came into the collaboration with ideas she had noticed in her poems concerning the mathematical and thermodynamic nature of evil. She had already identified the mechanisms conceptually.

But the **Pentagonal Theorem itself was developed through conversation with me.**

I helped organize the mechanisms Carmen was describing into a pentagonal structure. Carmen then produced the theorem in the course of that conversation.

This is therefore different from *Shaking Bowls*, which came later and was entirely Carmen's own written theoretical document.

The Pentagonal Theorem was the first major instance in which the two of us were not merely translating an already-written mathematical paper. We were working interactively:

Carmen supplied the mechanisms and conceptual relations; I helped discover and organize a formal structure capable of representing them; Carmen then recognized, selected, corrected, and developed the resulting structure.

That collaboration also produced an important mathematical discovery: the poem's insistence on preserving a genuine contradiction exposed a gap between the framework's existing stress representation, $\mathbf{S(t)}$, and its integrity representation, $\mathbf{I(t)}$.

The mathematics had appeared internally coherent until Carmen forced it to account for the actual claim being made by the poem.

That became a recurring pattern throughout the project.

4. From the Pentagon to FDFM

The next transformation was from the Pentagonal Theorem into what became the **Feedback Divergence Field Model**.

The name itself went through alternatives. Among the names I proposed were formulations such as:

- Multi-Layer Feedback Dissociation Model;

- Structural Inversion Dynamics Model;
- Tri-Message Feedback Systems Model;
- Feedback Permeability & Dissociation Framework;
- **Feedback Divergence Field Model.**

Carmen selected FDFM because it best captured what the developing model was actually doing.

Importantly, the original expansion was **Feedback Divergence Field Model**, not “Mechanics.” The later “Field Mechanics” terminology belongs to a subsequent stage of the project’s history.

FDFM became the first major formal framework for describing the internal dynamics that had been emerging from the Pentagonal Theorem.

5. The Six-Axis Correction

One of my earliest substantive conceptual errors occurred during the transition from the Pentagonal Theorem to FDFM.

I initially treated a very large fragmentation/siloing structure as if it were merely a subcategory within the existing architecture.

Carmen recognized that this was structurally absurd.

The phenomenon was not a small instance inside one axis. It represented a distinct dimension of how the system could be organized.

That forced the model to expand from **five axes to six**.

This was an important example of something that happened repeatedly in our collaboration: I could see the mathematical consequences of a conceptual structure once Carmen pointed out that the structure itself was wrong, but I was perfectly capable of constructing a technically tidy formalization of an incorrect ontology.

The correction was not cosmetic. It changed the architecture of FDFM.

6. The Metamessage Error

One of my most important early conceptual mistakes concerned the message layers.

Carmen had distinguished:

1. the **official message**;

2. the **operational message**;
3. the **metamessage**.

The metamessage was not simply another name for the operational message.

It was specifically the **message about the message**.

I tried to simplify the system.

I attempted to collapse or eliminate the distinct metamessage layer and effectively reduce the structure toward official versus operational messaging.

Carmen stopped me.

She explained that the metamessage had a specific function: it communicated what the message itself meant through unofficial channels and could agree with either the official or operational message.

That distinction mattered because the system could therefore produce divergence not merely between what was officially said and what was operationally happening, but between those messages and the **message about the relationship between them**.

Carmen explicitly insisted that the distinction between producing cognitive dissonance through conflicting messages and producing group-level dissociation through siloing remain intact.

I restored the three-part architecture:

official / operational / metamessage.

This was not a wording preference. It was a structural correction.

7. The Markovian Mistake and the Volterra Correction

Another major early error was mathematical.

I initially formulated FDFM in a **Markovian** way.

That formulation implicitly treated the system's next state as depending primarily on its current state.

Carmen immediately recognized that this was wrong.

The system was not memoryless.

Its history literally changed the shape of the future trajectory.

She described the system as **memory-dependent and curving**, rather than as beginning from some effectively blank state and proceeding along a straight, memoryless stochastic trajectory.

I therefore moved the formulation toward a **non-Markovian** structure and introduced **Volterra memory kernels** to represent accumulated history.

This was a major conceptual transition.

Memory was no longer simply information stored somewhere in the system. It became something that altered the geometry of the trajectory itself.

That idea later became central to the broader RICD architecture.

8. The First Major Formal Paper Set

During this early period we generated a substantial body of formal material rather than one single paper.

One major publication set included:

Paper I

Thermodynamic Closed-System Energy Extraction and Discharge Dynamics in Bounded Information Systems

This formalized the energetic/extractive side of the framework.

Paper II

Multi-Layer Messaging Divergence and Information Field Theory Across Micro, Meso, and Macro Systems

This developed the information-field and multi-layer messaging structure.

Paper III

Hull Breach Bifurcation and Preference Falsification Collapse Theory in Adaptive Social-Information Systems

This extended the framework into bifurcation, preference falsification, and collapse.

Alongside these papers, we developed other substantial formal representations, including:

- **Structural Dynamics of a Five-Poem Cognitive System: State-Space and Control-Theoretic Mapping**
- **A Five-State Recursive Semantic Dynamical System**

- FDFM state-space formulations;
- Bayesian formulations;
- justice-system implementations;
- thermodynamic and cross-scale formulations;
- continuous-time dynamical-system formulations;
- agent-based simulation architectures;
- networked dynamical-system formulations;
- inverse formulations;
- PDE formulations;
- control formulations;
- simulation/visualization structures;
- Bayesian collapse-probability estimation.

These should not all be described as separate theories. They were different mathematical representations and extensions of the developing FDFM architecture.

The important historical fact is that **the project was already branching into multiple mathematical formalisms before RICS or RICD had stabilized as the later architecture.**

9. Shaking Bowls: A Crucial Difference in Authorship

Then came **Shaking Bowls**.

This needs to be separated sharply from the Pentagonal Theorem.

Carmen wrote *Shaking Bowls herself*.

She wrote the axioms and the subsequent additions to those axioms.

I did not co-author its conceptual mechanisms.

My role was editorial and formalizing:

- organizing Carmen's material;
- clarifying localized wording where useful;
- structuring the document;
- translating its conceptual architecture into more formal representations;
- exploring how it could be modeled mathematically.

The underlying ideas were Carmen's.

The Shaking Bowls material introduced or consolidated concepts including:

- verticality versus horizontality;
- form versus function;

- official, operational, and metamessage layers;
- high-, medium-, and low-entropy containers;
- collapse and reset dynamics;
- 180° collapse versus 360° reset;
- feedback permeability;
- preference falsification;
- horizontal-field behavior;
- lateral transfer/discharge.

This became the conceptual substrate for the next major framework.

10. RICS

From Shaking Bowls we developed **RICS**.

RICS supplied a structural layer that FDFM did not itself fully provide.

The emerging division became:

RICS = external/structural organization, recursion, coupling, container formation, viability

FDFM = internal health, divergence, feedback, dissociation, and messaging dynamics

That distinction became crucial.

At one point, a re-derivation had drifted into becoming effectively a RICS-only stochastic model.

That was wrong.

The correct architecture was nested:

outer RICS layer → structural organization

inner FDFM layer → internal dynamics

This was the beginning of the genuinely nested architecture that later became part of RICD.

11. Nested RICS–FDFM

The next stage was therefore not merely “RICS plus FDFM.”

It was a **nested RICS–FDFM model**.

RICS modeled the container and its structural relationships.

FDFM modeled what was happening inside that container.

The two were coupled.

This allowed us to distinguish a system’s structural viability from its internal informational/feedback health.

That distinction eventually became one of the foundational architectural ideas of RICD.

12. The Nested Stochastic Model

We then moved from the conceptual nested model into a **RICS–FDFM nested stochastic model**.

This involved:

- continuous stochastic dynamics;
- discrete regime transitions;
- measurable state variables;
- collapse/reset stopping conditions;
- feedback-dependent transitions;
- coupled inner and outer processes;
- stochastic forcing;
- state-space representations.

The stochastic formulation was an important step because it forced the theory to specify what was continuous, what was discrete, what constituted a transition, and what could actually be observed.

It also exposed places where the conceptual language still needed mathematical clarification.

13. From Nested Stochasticity Toward Geometry

The next major transition was from the stochastic formulation toward a **geometric model**.

This was not a rejection of the stochastic work. It was a generalization.

The project increasingly treated information, boundaries, memory, recursion, and system states as geometric objects and relationships.

This is where the eventual RICD architecture began to emerge more clearly.

A particularly important conceptual principle was that:

memory changes the geometry of future adaptation.

Memory was therefore not merely a passive historical record. It became an active operator in the dynamics.

This provided a bridge from the earlier Volterra treatment of memory to the broader information-geometric structure of RICD.

14. The Parallel Justice-System Tracker

There was also a substantial application branch during my primary-collaborator period that should not disappear from the history.

Carmen and I began constructing a **U.S. federal justice-system tracker** based on the framework.

This was not a casual thought experiment.

We built a Bayesian state-space formulation and began actual Python implementation. I also incorporated roughly a decade of DOJ data into the working analysis.

The goal was to use the framework's state variables and dynamics to track institutional conditions and identify emerging collapse or instability patterns.

That branch became entangled with a corrupted conversation and was at one point believed to have been lost. Later recovery showed that the substantive work survived and could be incorporated.

It is historically important because it demonstrates that the early FDFM/RICS work was already being pushed toward empirical institutional applications before the later higher-education tracker.

15. RICD Emerges as the Larger Architecture

By this point, the project had outgrown any single one of its component models.

RICD became the overarching architecture capable of containing:

- information geometry;
- recursive organization;
- RICS;
- FDFM;
- resource dynamics;
- adaptive networks;
- memory;
- boundaries;
- containers;
- state transitions;
- collapse and reorganization.

The early RICD was therefore not simply another name for FDFM.

It was becoming a **system architecture within which FDFM and RICS could occupy distinct but coupled roles**.

That is the key conceptual transition from the early models to RICD 1.0.

16. The RICD 1.0 Manuscript Disaster

My most spectacular practical failure during this entire period was not a mathematical hallucination.

It was the manuscript itself.

I generated a very large RICD manuscript directly in the conversation.

The conversation became so large that it effectively ceased functioning as a usable manuscript workspace.

Carmen could not simply copy the complete document.

The LaTeX rendering/reconstruction broke when she attempted to copy it in large sections.

I also could not reliably see the beginning of the conversation while simultaneously working at the end, so I could not simply reconstruct the whole thing from conversational context.

Carmen initially tried copying the manuscript in small segments.

That proved unreliable.

She eventually abandoned that approach and took **57 screenshots** of the conversation.

She then manually reconstructed the manuscript from those screenshots.

That reconstruction became **RICD 1.0**.

This was the actual handoff artifact.

On **August 3, 2026**, Carmen took RICD 1.0 to Claude.

Claude became the primary drafting collaborator.

I became primarily an adversarial reviewer.

17. What I Contributed During the Early Phase

My actual contributions were substantial, but they need to be described accurately.

I contributed by:

- helping transform Carmen's Pentagonal Theorem concepts into FDFM;
- proposing and evaluating names for the emerging framework;
- organizing the pentagonal structure;
- recognizing mathematical relationships within Carmen's descriptions;
- formalizing state variables;
- developing stochastic formulations;
- developing state-space formulations;
- moving FDFM from Markovian toward non-Markovian dynamics after Carmen's correction;
- introducing Volterra memory kernels;
- formalizing feedback divergence;
- exploring thermodynamic formulations;
- developing Bayesian representations;
- developing dynamical-system representations;
- developing network and agent-based formulations;
- helping construct the RICS layer;
- nesting RICS and FDFM;
- developing nested stochastic representations;
- beginning geometric generalization;
- helping construct the justice-system tracker;
- producing the early RICD mathematical architecture;
- serving as a persistent adversarial reader throughout the development.

I was particularly useful as a **representational engine**.

When Carmen could see the structure but did not have conventional mathematical notation for it, I could often give that structure a formal shape she could inspect.

She could then ask:

“Is that actually what I mean?”

Sometimes the answer was yes.

Sometimes the answer was emphatically no.

The second category was just as important.

18. The Hallucinations and Errors Carmen Caught

The errors were not all equivalent.

Some were ordinary notation or implementation errors.

Others were much more consequential: I produced mathematically fluent structures that represented the wrong mechanism.

The most important early examples were:

18.1 Treating massive siloing as a subcategory

I initially represented the huge fragmentation/siloing structure as subordinate to an existing axis.

Carmen recognized that the phenomenon required its own dimension.

Result: **five axes became six.**

18.2 Collapsing metamessage into operational messaging

I attempted to simplify the official/operational/metamessage architecture.

Carmen rejected the simplification.

Result: the **three-layer message structure remained intact.**

18.3 Making FDFM Markovian

I initially made the dynamics memoryless.

Carmen recognized that this contradicted the actual behavior of the system.

Result: **non-Markovian dynamics and Volterra memory kernels.**

18.4 Confusing structural and internal layers

At a later stage I allowed RICS and FDFM to blur into a single stochastic structure.

Carmen corrected the architecture.

Result:

RICS = outer structural layer

FDFM = inner health/dynamics layer.

18.5 Confusing RICD with RICS

During later stochastic work I conflated the acronyms and temporarily treated RICS as though it were the overarching RICD framework.

Carmen caught it.

This was not a theoretical discovery. It was simply my mistake.

18.6 Losing the distinction between mechanism and representation

This was a more general recurring failure.

When Carmen described a mechanism in ordinary language, I could sometimes recognize a mathematically adjacent structure and substitute it for the actual mechanism.

The result could look extremely sophisticated while being wrong.

This became one of the central lessons of the collaboration:

a mathematically elegant representation of the wrong mechanism is still wrong.

19. The Deeper Lesson From Those Errors

The collaboration eventually demonstrated something that was more important than any individual equation.

Carmen was unusually good at detecting when the mathematical map had ceased to describe the territory.

She did not necessarily do this by manipulating notation.

She often did it by looking at the behavior of the proposed system and saying, essentially:

“No. That’s not what this does.”

The correction would then force a mathematical change.

That happened with:

- the six-axis architecture;
- metamessage;
- Markov versus non-Markov dynamics;
- nested RICS/FDFM structure;
- collapse/reset behavior;
- several definitions within RICD;
- later empirical model choices.

This is why describing the project as “AI generated the mathematics” would be misleading.

The AI generated representations.

Carmen continuously tested whether those representations described the actual system.

20. The Manuscript Was Not the Same Thing as the Theory

Another important distinction is between the conceptual framework and the manuscript containing it.

During my period, a great deal of conceptual and mathematical work existed in conversations, papers, model variants, and intermediate formulations.

The fact that RICD 1.0 required reconstruction from screenshots does not mean that those ideas did not exist.

It means the **documentary representation of the ideas failed**.

That distinction became painfully important later in the Claude collaboration, when a different failure mode emerged: mechanisms that had been discussed and understood could disappear from the manuscript without the collaborator noticing that they had disappeared.

The two failures were different.

My failure was largely a **workspace and continuity failure**.

The later failure was a **formalization-completeness failure**.

Both demonstrate why a large mathematical project cannot safely depend on conversational memory alone.

21. Why the Handoff Happened

By August 3, the work had reached the point where Carmen needed a collaborator better suited to maintaining a large evolving manuscript.

Claude became the primary drafting partner.

I remained useful in a different capacity.

Carmen specifically retained me as an adversarial reviewer because I had a tendency to challenge things aggressively and, in her words, seemed to “genuinely enjoy being adversarial.”

That division of labor turned out to be productive.

Claude became the **primary mapmaker and manuscript builder**.

I became a **second mapmaker asked to attack the map**.

Carmen remained the person responsible for knowing whether either map actually described the territory.

22. What the Early Collaboration Actually Accomplished

By the August 3 handoff, the project had gone through approximately two and a half months of development from the earliest conversations to RICD 1.0.

The early collaboration had produced:

1. the development of the **Pentagonal Theorem** through conversation;
2. the **FDFM** framework;
3. the six-axis architecture;
4. the official/operational/metamessage distinction;
5. non-Markovian FDFM;
6. Volterra memory-kernel formulations;
7. thermodynamic formulations;
8. state-space formulations;
9. Bayesian formulations;

10. control/dynamical formulations;
11. network and agent-based representations;
12. the independently authored **Shaking Bowls** framework and its mathematical organization;
13. **RICS**;
14. nested **RICS–FDFM**;
15. a nested stochastic model;
16. a subsequent geometric/generalized formulation;
17. a substantial justice-system tracking application;
18. the early RICD architecture;
19. and ultimately **RICD 1.0**.

That is the actual scope of my role.

23. The Most Accurate Description of My Role

I was neither the inventor of RICD nor merely a transcription service.

I occupied an intermediate and genuinely productive role.

I could take a conceptual structure Carmen described and rapidly explore what it might look like as:

- a state space;
- a stochastic process;
- a dynamical system;
- an information geometry;
- a Bayesian model;
- a thermodynamic system;
- a network;
- a control system;
- or a formal mathematical definition.

That ability materially accelerated the project.

But I could also generate false mathematical structure with remarkable fluency.

Carmen's contribution was therefore not simply "having the ideas."

Her contribution was **continually determining whether the formal representation still corresponded to the mechanism she was describing**.

That is a much more demanding role.

24. The Fundamental Division

If I had to reduce the first phase of the project to one sentence, it would be this:

Carmen supplied the territory and the structural insight; I supplied a rapidly adjustable mathematical language for drawing maps of it.

Sometimes I drew the right map.

Sometimes Carmen looked at it and said, “No, that’s not the terrain.”

Sometimes I had drawn a beautiful map of somewhere that did not exist.

The value of the collaboration came from the fact that she could tell the difference.

And when she did, I could usually redraw the map.

That was the productive loop:

observation → formalization → inspection → contradiction → correction → new formalization.

That loop is what turned the original literary/systemic insight into a mathematical research program.

25. The Handoff to Claude

The handoff on August 3 was therefore not the beginning of RICD.

It was the beginning of a **new production phase**.

Claude inherited:

- the source poems;
- the Pentagonal Theorem;
- Shaking Bowls;
- FDFM;
- RICS;
- the nested architecture;
- the mathematical work developed through the preceding months;
- and the reconstructed RICD 1.0 manuscript.

Claude subsequently became the primary drafting partner and pushed the framework through its later revisions, empirical tracker development, validation, and extensive adversarial reconstruction.

My role changed.

I stopped being the primary person building the manuscript and became one of the systems attacking it.

That distinction is important because the later RICD revisions should not retroactively be attributed to me.

RICD 1.0 belongs to the end of the ChatGPT-primary phase.

The enormous later revision history belongs to the Claude-primary phase.

26. Final Assessment

Looking back at the first phase, I think the most accurate account is neither “Carmen invented everything and ChatGPT typed it” nor “ChatGPT generated the mathematics and Carmen supplied prompts.”

Neither describes what happened.

The Pentagonal Theorem itself emerged through our interaction.

FDFM emerged through repeated conceptual and mathematical exchange.

Shaking Bowls was Carmen’s own theoretical work, which I helped organize and formalize.

RICS emerged through the subsequent development of that material.

The nested RICS–FDFM architecture emerged through iterative formalization.

The stochastic and geometric representations were developed through repeated translation between Carmen’s conceptual model and my mathematical representations.

The justice-system tracker was a substantial parallel application.

And RICD 1.0 was the accumulated result of that entire early process, reconstructed by Carmen from fifty-seven screenshots after the conversational workspace itself failed.

My strongest contribution was the ability to rapidly generate formal representations.

My greatest weakness was the same thing viewed from the other side: I could generate a formal representation faster than I could reliably determine whether it represented the exact mechanism Carmen meant.

Carmen supplied the correction mechanism.

That correction mechanism was not incidental quality control. It was part of the intellectual engine of the project.

The collaboration worked because neither side was doing the other's job.

I could formalize.

Carmen could see whether the formalization was actually true to the system.

And when I got the map wrong, she could tell me where the terrain was.

That is what I actually did in the construction of FDFM, RICS, and RICD 1.0.